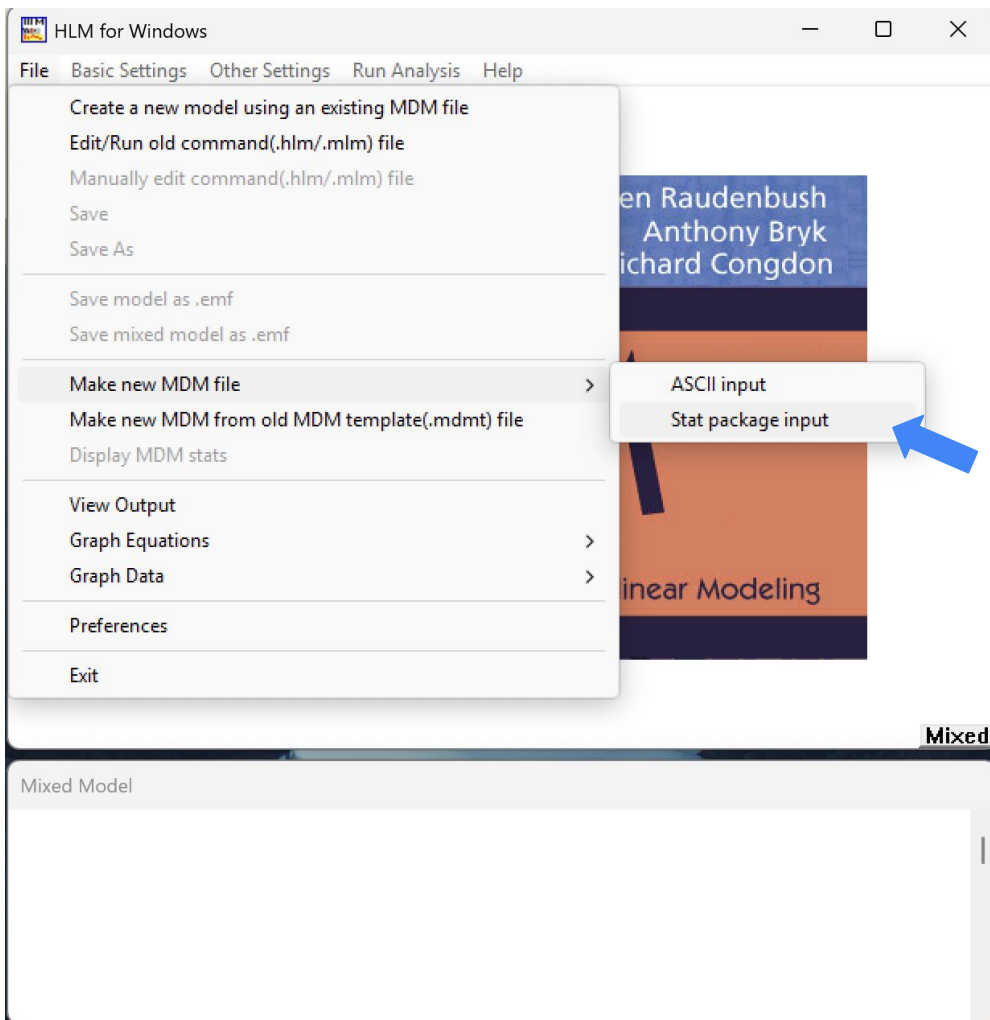


Basics of Running HLM



Select MDM type

Nested Models

☒ HLM2 ☐ HLM3 ☐ HLM4



Hierarchical Multivariate Linear Models

☐ HMLM ☐ HMLM2

Cross-classified Models

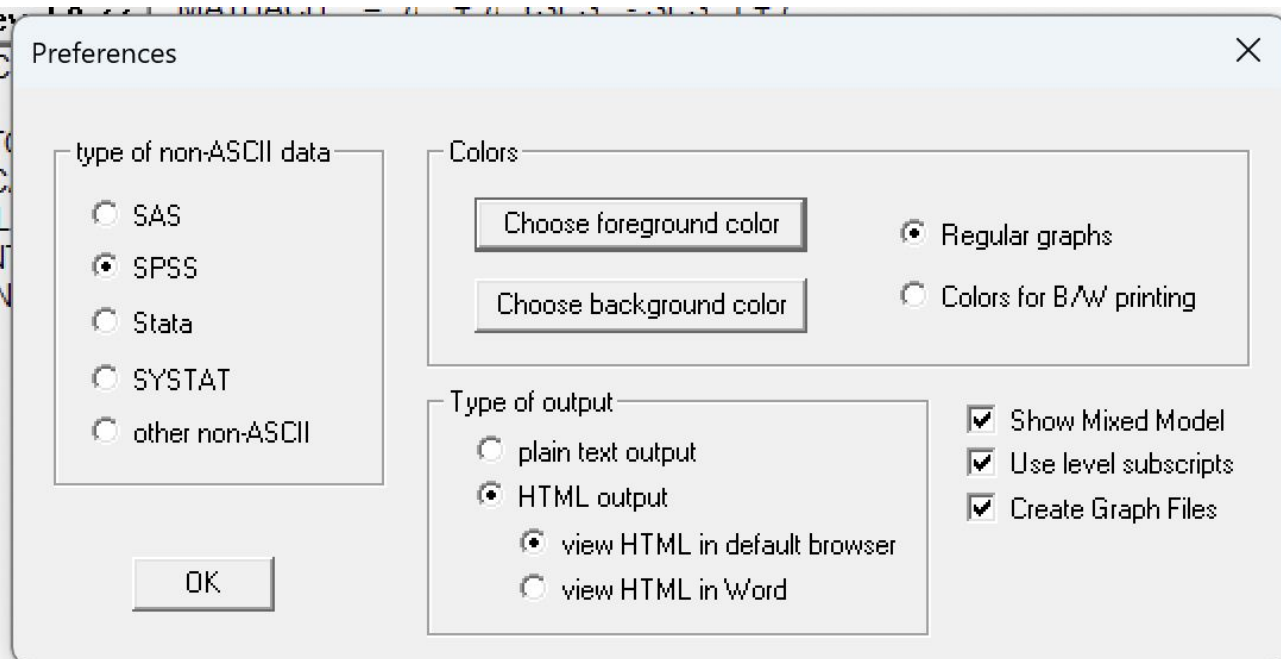
☐ HCM2 ☐ HLM-HCM ☐ HCM3

OK

Cancel

For today, we are trying
a 2 level HLM... HLM2

Check your preferences (File → Preferences) to make sure to Show Mixed Model!



The image shows the 'Preferences' dialog box in SPSS. The 'type of non-ASCII data' section on the left has 'SPSS' selected. The 'Colors' section on the right has 'Regular graphs' selected. The 'Type of output' section at the bottom left has 'HTML output' selected, with 'view HTML in default browser' also selected. The 'Show Mixed Model', 'Use level subscripts', and 'Create Graph Files' checkboxes are all checked. An 'OK' button is at the bottom left.

Preferences

type of non-ASCII data

- ☐ SAS
- ☒ SPSS
- ☐ Stata
- ☐ SYSTAT
- ☐ other non-ASCII

OK

Colors

Choose foreground color

Choose background color

- ☒ Regular graphs
- ☐ Colors for B/W printing

Type of output

- ☐ plain text output
- ☒ HTML output
 - ☒ view HTML in default browser
 - ☐ view HTML in Word

- ☒ Show Mixed Model
- ☒ Use level subscripts
- ☒ Create Graph Files

HLM for Windows

File Basic Settings Other Settings Run Analysis Help

Make MDM - HLM2

MDM template file

File Name:

Open mdmt file Save mdmt file Edit mdmt file

MDM File Name (use .mdm suffix)

Trial 2.01

Input File Type SPSS/Windows

Structure of Data - this affects the notation only!

☒ cross sectional (persons within groups) ☐ measures within groups

☐ longitudinal (occasions within persons)

Level-1 Specification

Browse Level-1 File Name: C:\Mac\Home\Documents\HLMData\HSB1.SAV Choose Variables

Missing Data? Delete missing level-1 data when:

☒ No ☐ Yes ☐ making mdm ☐ running analyses

Level-2 Specification

Browse Level-2 File Name: C:\Mac\Home\Documents\HLMData\HSB2.SAV Choose Variables

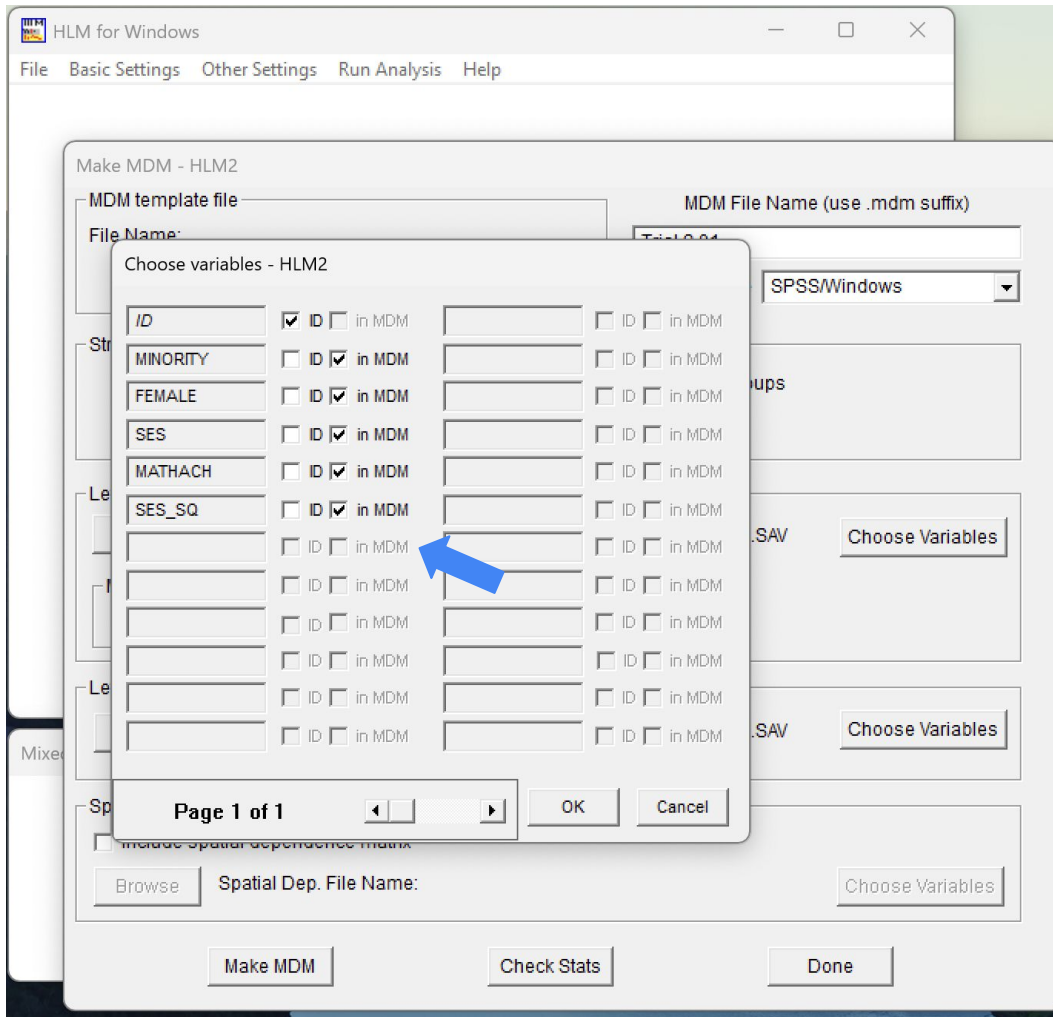
Spatial Dependence Specification

☐ Include spatial dependence matrix

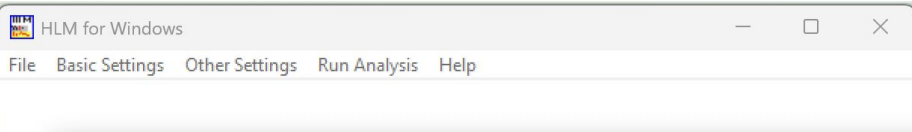
Browse Spatial Dep. File Name: Choose Variables

Make MDM Check Stats Done

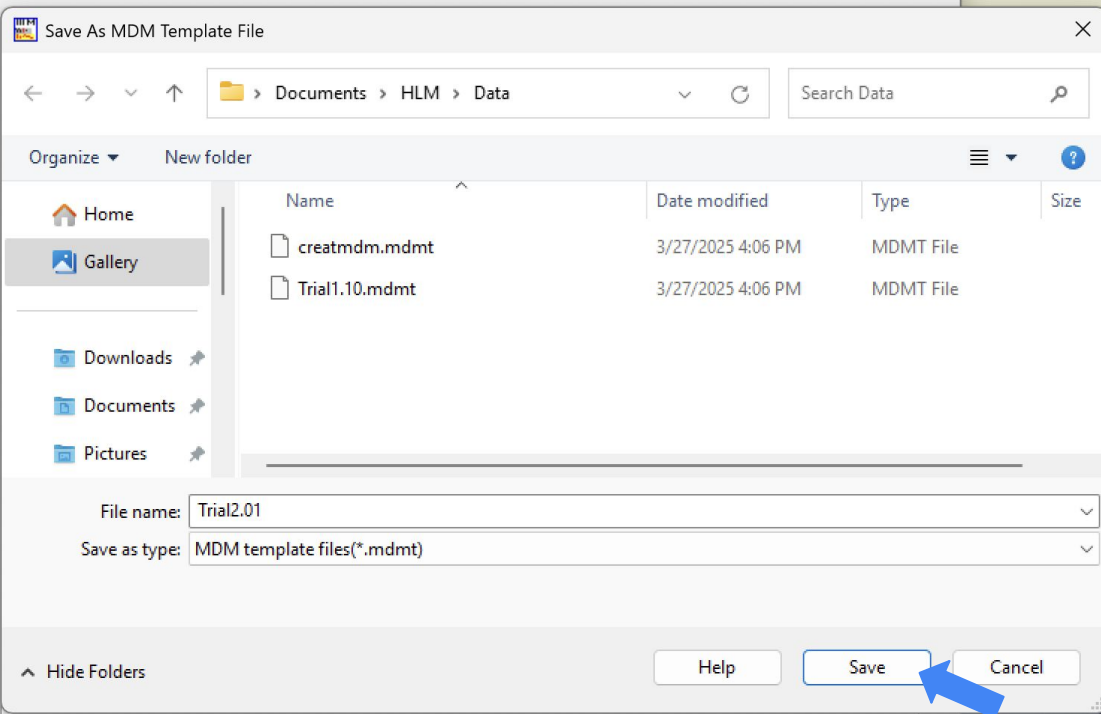
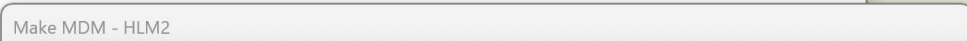
Upload data for both Level 1 and Level 2 following the same steps. Put in data, choose variables.



Select ID for your student ID level and in Model for the remaining variables.



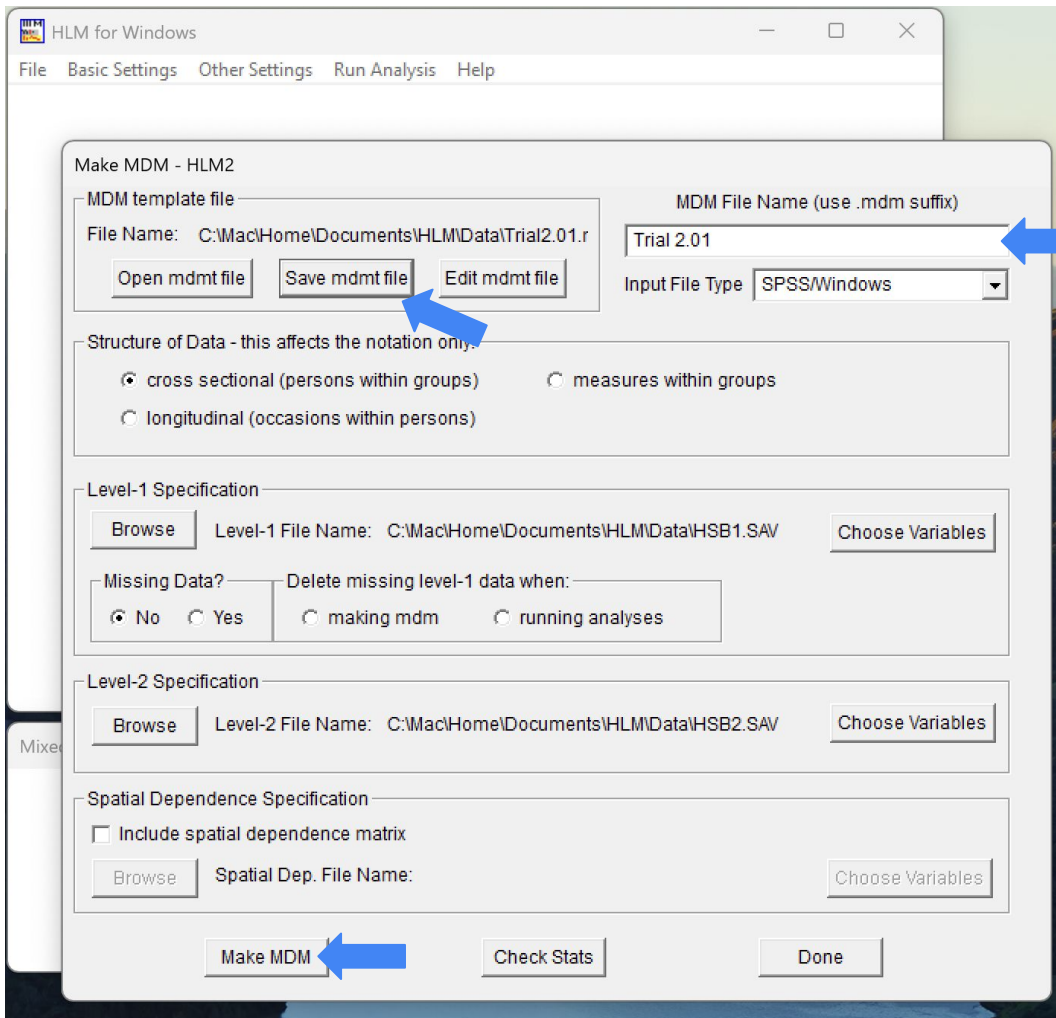
Save file



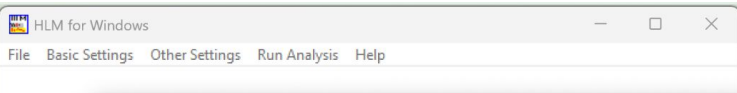
Make MDM

Check Stats

Done



Save mdmt file



First output is
Descriptive statistics.
Close with X at top
corner when done.

MDM2MDM.STS

File Edit View

LEVEL-1 DESCRIPTIVE STATISTICS

VARIABLE NAME	N	MEAN	SD	MINIMUM	MAXIMUM
MINORITY	7185	0.27	0.45	0.00	1.00
FEMALE	7185	0.53	0.50	0.00	1.00
SES	7185	0.00	0.78	-3.76	2.69
MATHACH	7185	12.75	6.88	-2.83	24.99
SES_SQ	7185	0.61	0.77	0.00	14.12

LEVEL-2 DESCRIPTIVE STATISTICS

VARIABLE NAME	N	MEAN	SD	MINIMUM	MAXIMUM
SIZE	160	1097.83	629.51	100.00	2713.00
SECTOR	160	0.44	0.50	0.00	1.00
PRACAD	160	0.51	0.26	0.00	1.00
DISCLIM	160	-0.02	0.98	-2.42	2.76
HIMINTY	160	0.28	0.45	0.00	1.00
MEANSES	160	-0.00	0.41	-1.19	0.83

MDM template: C:\Mac\Home\Documents\HLM\Data\Trial2.01.mdm
MDM file name: Trial 2.01
Date: Mar 28, 2025
Time: 07:31:30

Ln 1, Col 1 1,202 characters 100% Windows (CRLF) UTF-8

HLM for Windows

File Basic Settings Other Settings Run Analysis Help

Make MDM - HLM2

MDM template file

File Name: C:\Mac\Home\Documents\HLM\Data\Trial2.01.r

Open mdmt file Save mdmt file Edit mdmt file

MDM File Name (use .mdm suffix)

Trial 2.01

Input File Type SPSS/Windows

Structure of Data - this affects the notation only!

☒ cross sectional (persons within groups) ☐ measures within groups

☐ longitudinal (occasions within persons)

Level-1 Specification

Browse Level-1 File Name: C:\Mac\Home\Documents\HLM\Data\HSB1.SAV Choose Variables

Missing Data? Delete missing level-1 data when:

☒ No ☐ Yes ☐ making mdm ☐ running analyses

Level-2 Specification

Browse Level-2 File Name: C:\Mac\Home\Documents\HLM\Data\HSB2.SAV Choose Variables

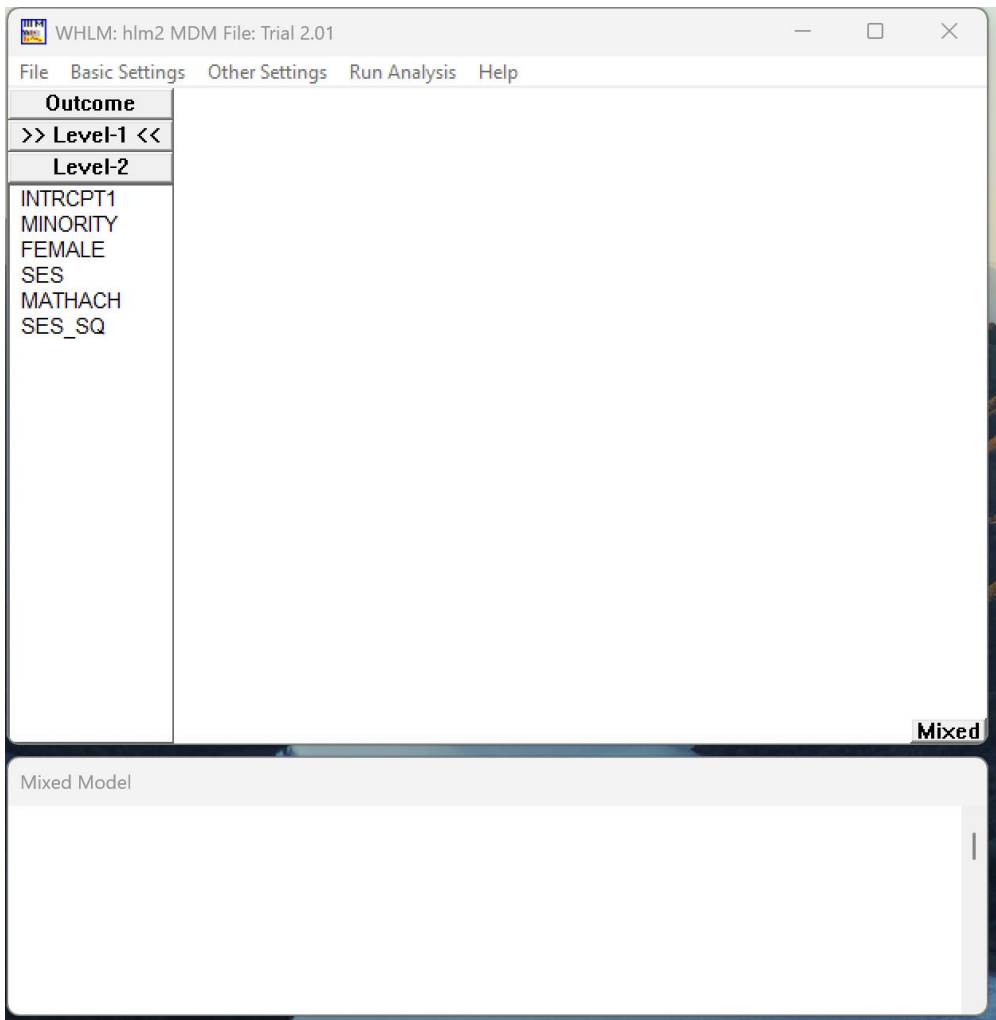
Spatial Dependence Specification

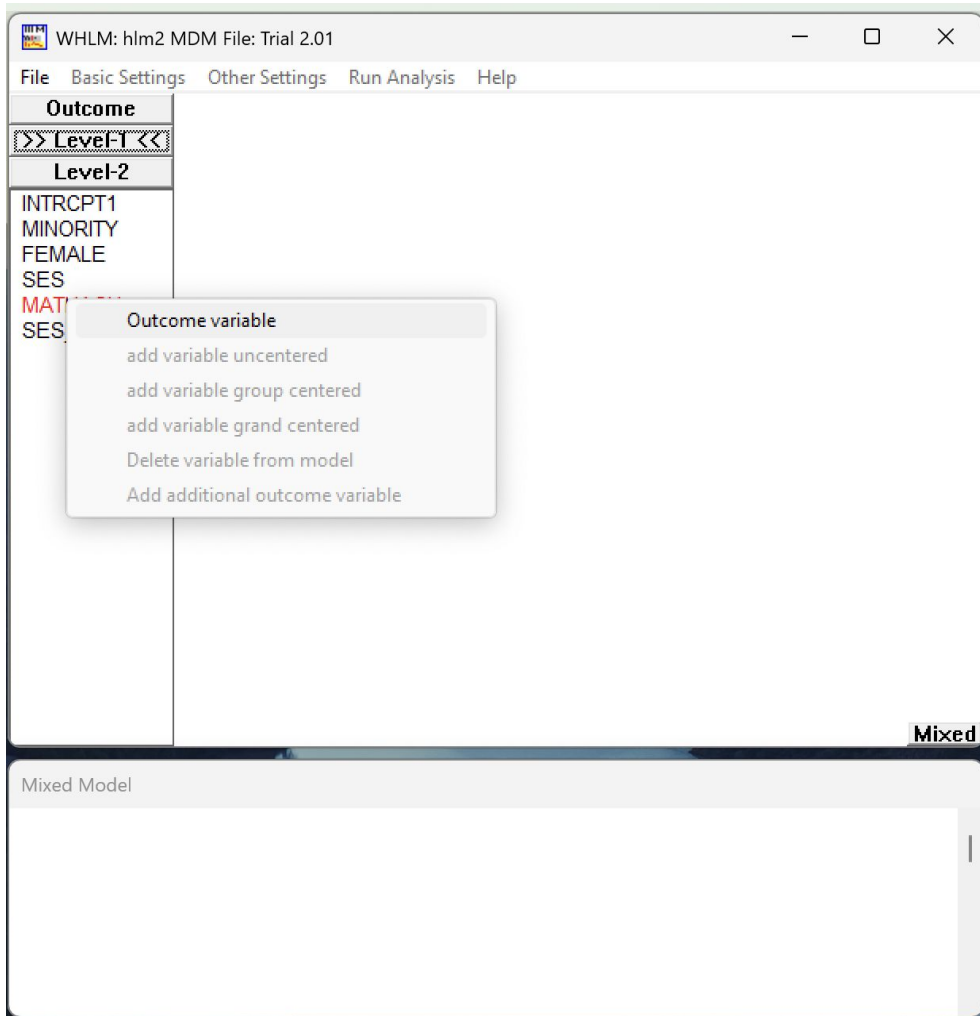
☐ Include spatial dependence matrix

Browse Spatial Dep. File Name: Choose Variables

Make MDM Check Stats Done

To run model, click
done.





Add outcome variable
based on your model
and research
questions.

WHLM: hlm2 MDM File: Trial 2.01

File Basic Settings Other Settings Run Analysis Help

Outcome

>> Level-1 <<

Level-2

INTRCPT1
MINORITY
FEMALE
SES
MATH
SES_

LEVEL 1 MODEL

$$\text{MATHACH}_{ij} = \beta_{0j} + r_{ij}$$

LEVEL 2 MODEL

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

Mixed

Mixed Model

$$\text{MATHACH}_{ij} = \gamma_{00} + u_{0j} + r_{ij}$$

- Outcome variable
- add variable uncentered
- add variable group centered
- add variable grand centered
- Delete variable from model
- Add additional outcome variable

Other variables can be added as centered group (within) or grand (between).

WHLM: hlm2 MDM File: Trial 2.01

File Basic Settings Other Settings Run Analysis Help

Outcome

Level-1

Level-2

LEVEL 1 MODEL

$$\text{MATHACH}_{ij} = \beta_{0j} + \beta_{1j}(\text{SES}_{ij} - \overline{\text{SES}}_{.j}) + r_{ij}$$

LEVEL 2 MODEL

add variable uncentered

add variable grand centered

Delete variable from model

Add as additional outcome variable

Mixed

Mixed Model

$$\text{MATHACH}_{ij} = \gamma_{00} + \gamma_{01}*(\text{SECTOR}_j - \overline{\text{SECTOR}}_{.j}) + \gamma_{10}*(\text{SES}_{ij} - \overline{\text{SES}}_{.j}) + u_{0j} + r_{ij}$$

Add outcome variable based on your model and research questions. Repeat for level 2. Use grand for level 2.

WHLM: hlm2 MDM File: Trial 2.01

File Basic Settings Other Settings Run Analysis Help

Outcome

Level-1

>> Level-2 <<

INTRCPT2
SIZE
SECTOR
PRACAD
DISCLIM
HIMINTY
MEANSES

LEVEL 1 MODEL

$$\text{MATHACH}_{ij} = \beta_{0j} + \beta_{1j}(\text{SES}_{ij} - \overline{\text{SES}}_{\cdot j}) + r_{ij}$$

LEVEL 2 MODEL

$$\beta_{0j} = \gamma_{00} + \gamma_{01}(\text{SECTOR}_j - \overline{\text{SECTOR}}_{\cdot}) + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + u_{1j}$$

Mixed

Mixed Model

$$\text{MATHACH}_{ij} = \gamma_{00} + \gamma_{01}(\text{SECTOR}_j - \overline{\text{SECTOR}}_{\cdot}) + \gamma_{10}(\text{SES}_{ij} - \overline{\text{SES}}_{\cdot j}) + u_{0j} + r_{ij}$$

WHLM: hlm2 MDM File: Trial 2.01

File Basic Settings Other Settings Run Analysis Help

Outcome

Level-1

>> Level-2 <<

INTRCPT2
SIZE
SECTOR
PRACAD
DISCLIM
HIMINTY
MEANSES

LEVEL 1 MODEL

$$\text{MATHACH}_{ij} = \beta_{0j} + \beta_{1j}(\text{SES}_{ij} - \overline{\text{SES}}_{\cdot j}) + r_{ij}$$

LEVEL 2 MODEL

$$\beta_{0j} = \gamma_{00} + \gamma_{01}(\text{SECTOR}_j - \overline{\text{SECTOR}}_{\cdot}) + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + u_{1j}$$

Mixed

Mixed Model

$$\text{MATHACH}_{ij} = \gamma_{00} + \gamma_{01}(\text{SECTOR}_j - \overline{\text{SECTOR}}_{\cdot}) + \gamma_{10}(\text{SES}_{ij} - \overline{\text{SES}}_{\cdot j}) + u_{0j} + u_{1j}(\text{SES}_{ij} - \overline{\text{SES}}_{\cdot j}) + r_{ij}$$

Be sure to click on the model variables you want to include. Greyed out do not get included in the mixed model.

WHLM: hlm2 MDM File: Trial 2.01

File Basic Settings Other Settings Run Analysis Help

- Create a new model using an existing MDM file
- Edit/Run old command(.hlm/.mlm) file
- Manually edit command(.hlm/.mlm) file
- Save
- Save As
- Save model as .emf
- Save mixed model as .emf
- Make new MDM file >
- Make new MDM from old MDM template(.mdmt) file
- Display MDM stats
- View Output
- Graph Equations >
- Graph Data >
- Preferences
- Exit

Mixed

Mixed Model

$$\text{MATHACH}_{ij} = \gamma_{00} + \gamma_{01} * (\text{SECTOR}_j - \overline{\text{SECTOR}_.}) + \gamma_{10} * (\text{SES}_{ij} - \overline{\text{SES}_{.j}}) + u_{0j} + u_{1j} * (\text{SES}_{ij} - \overline{\text{SES}_{.j}}) + r_{ij}$$

View output.

WHLM: hlm2 MDM File: Trial 2.01

File Basic Settings Other Settings Run Analysis Help

Outcome

Level-1

>> **Level-2** <<

INTRCPT2
SIZE
SECTOR
PRACAD
DISCLIM
HIMINTY
MEANSES

LEVEL 1 MODEL

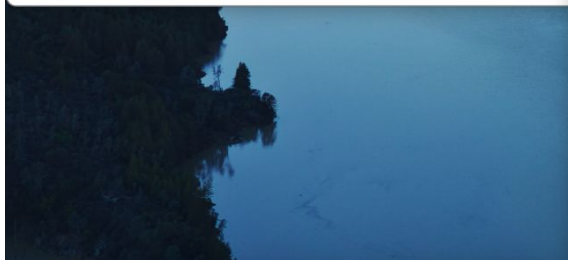
$$\text{MATHACH}_{ij} = \beta_{0j} + \beta_{1j}(\text{SES}_{ij} - \overline{\text{SES}}_{\cdot j}) + r_{ij}$$

LEVEL 2 MODEL

$$\beta_{0j} = \gamma_{00} + \gamma_{01}(\text{SECTOR}_j - \overline{\text{SECTOR}}_{\cdot}) + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + u_{1j}$$

Mixed Model

$$\text{MATHACH}_{ij} = \gamma_{00} + \gamma_{01}(\text{SECTOR}_j - \overline{\text{SECTOR}}_{\cdot}) + \gamma_{10}(\text{SES}_{ij} - \overline{\text{SES}}_{\cdot j}) + u_{0j} + r_{ij}$$


hlm2.html hlm2.html hlm2.html

File C:/Mac/Home/Documents/HLM/Data/hlm2.html

Module: HLM2 (7.30)
Date: Mar 27, 2025
Time: 16:19:10

Specifications for this HLM2 run

Problem Title: no title

The data source for this run = Lab1.1
The command file for this run = C:\Mac\Home\Documents\HLM\Data\Trial1.12.hlm
Output file name = C:\Mac\Home\Documents\HLM\Data\hlm2.html
The maximum number of level-1 units = 7185
The maximum number of level-2 units = 160
The maximum number of iterations = 100
Method of estimation: restricted maximum likelihood

The outcome variable is MATHACH

Summary of the model specified

Step 2 model

Level-1 Model

$$\text{MATHACH}_{ij} = \beta_{0j} + \beta_{1j}(\text{SES}_{ij}) + r_{ij}$$

Level-2 Model

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + u_{1j}$$

SES has been centered around the group mean.

Mixed Model

Interpret Results!